

DRUGS ACTING ON THE KIDNEY

Objectives

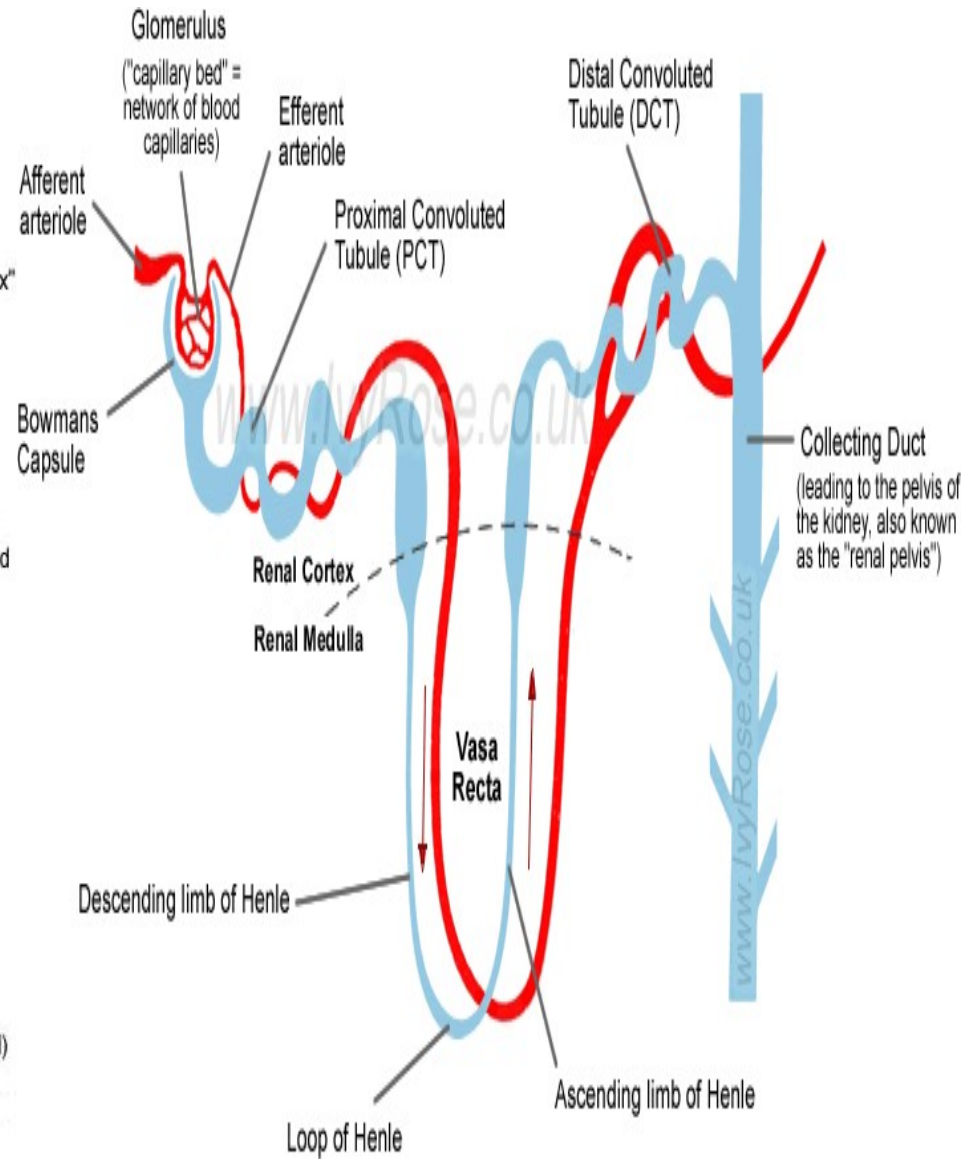
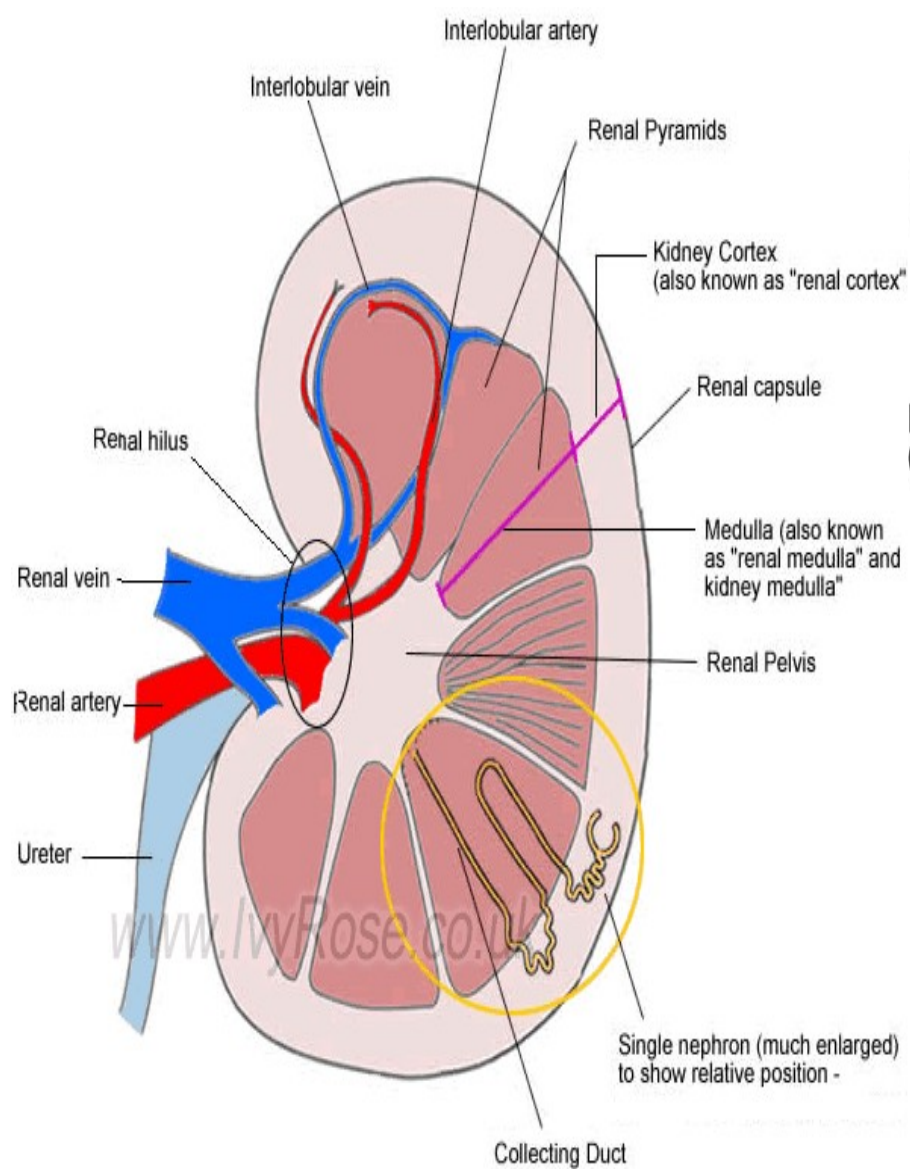
- **Up on completion of this chapter, the student will be able to:**
- Explain the physiology of urine formation
 - Elucidate the principles of diuretic action
 - Describe the pharmacological actions, mechanism of actions, adverse reactions and pharmacokinetics of the different diuretics.
 - List the therapeutic uses of those agents
 - Explain vasopressin and other agents affecting renal conservation of water

Contents

- Definition
- Urine formation
- Renal Tubular Transport Processes
- Principles of Diuretic Action
- Classification
 - Diuretics: Carbonic Anhydrase Inhibitors, Loop Diuretics, Thiazide Diuretics, Potassium Sparing Diuretics and Osmotic Diuretics
- Vasopressin and Other Agents

Diuretics

- Renal Physiology: the Physiology of urine formation
- Nephron-
 - ✓ Basic urine-forming unit of the kidney
 - ✓ Consists of **a filtering apparatus**, the glomerulus, connected to **a long tubular portion**
 - ✓ Each human kidney is composed of approximately one million nephrons.



- Urine formation involves 3 processes
 - 1) Glomerular Filtration,
 - 2) Tubular Reabsorption and
 - 3) Tubular Secretion in nephrons.

1. Glomerular Filtration

- As **blood flows through the kidney**, it passes into glomerular capillaries
- **All soluble components of blood** except plasma **protein** and **lipids** are filtered
- Around **180 liter of fluid filtered at glomerulus per day**.
- Glomerular capillary **hydrostatic pressure** drives (filters) water and electrolytes into Bowman's space and into PCT.

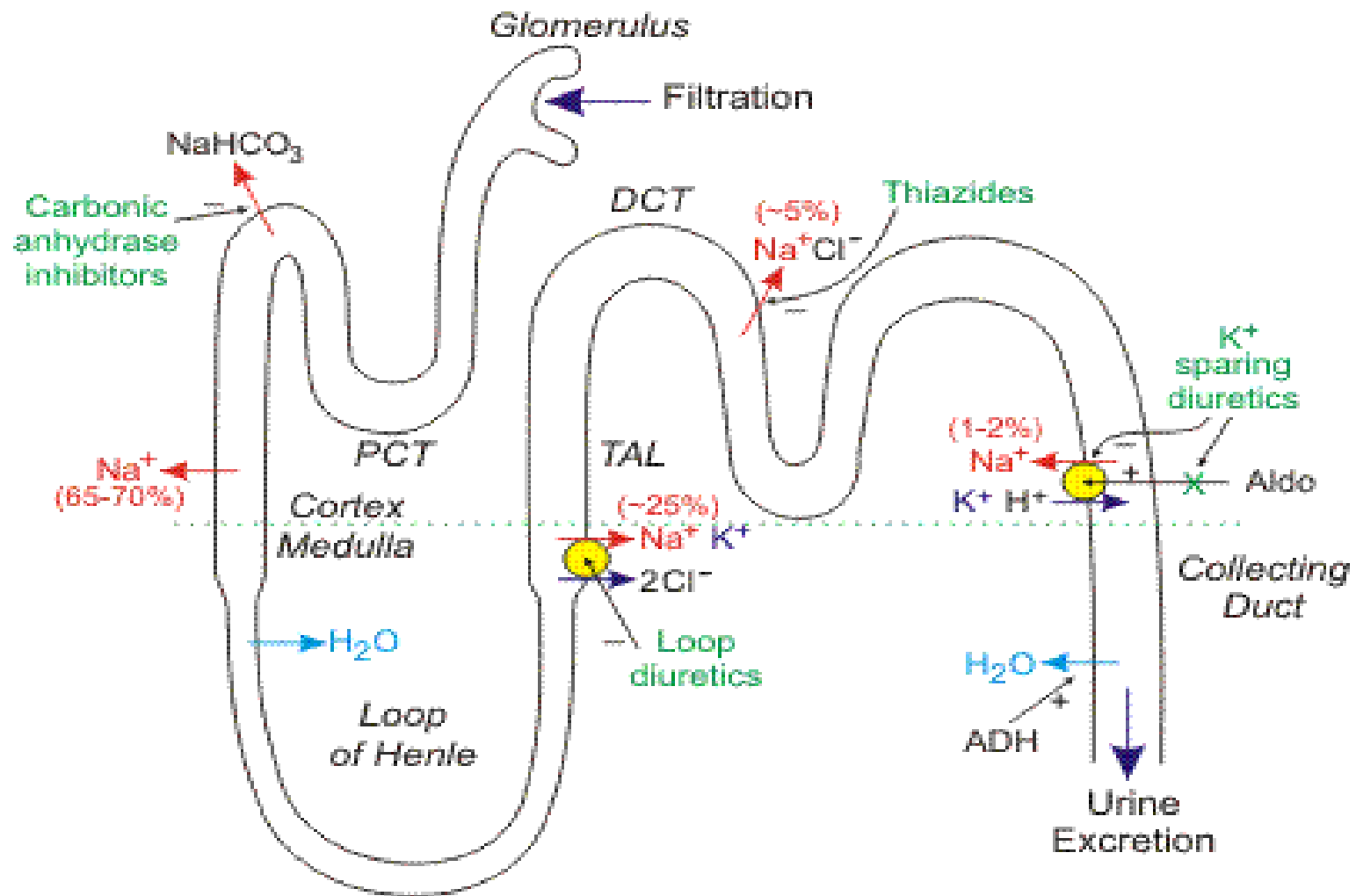
- **Solutes of small size flow with filtered water (solvent drag)**
- Formed **elements** and **macromolecules** are retained by the filtration barrier.

2. Tubular reabsorption

- After filtration, those substances that the body must conserve will be reabsorbed
- The two kidneys produce together approximately **120 ml/min of ultrafiltrate**,
- Yet only **1 ml/min (1.5L/day) of urine** is produced.
- Therefore, greater than 99% of the glomerular ultrafiltrate is reabsorbed
- Where does reabsorption takes place??
 - Occurs at four different sites with in a nephron

Proximal tubule

- Lies within the cortex and is attached with Bowman's capsule.
- The site of **sodium, water and bicarbonate transport**.
- **65-70% of the filtered sodium is** reabsorbed.
- This reabsorption is essentially isotonic.
 - B/c every molecule of sodium that is reabsorbed is accompanied by a molecule of water.



- This loop is consists of the **descending thin limb (DTL)**, **ascending thin limb (ATL)** and **the thick ascending limb (TAL)**.
- Approximately 25% of filtered Na^+ is reabsorbed mostly in the TAL, which has a large re-absorptive capacity.
- **DTL** is highly **permeable to water**, yet its permeability to **NaCl and urea** is low

- **ATL** is permeable to **NaCl** and **urea** but is **impermeable to water**.
- **TAL** actively reabsorbs **NaCl** but is **impermeable to water and urea**
 - ✓ Contributes to hypertonicity of medullary interstitium

Distal convoluted tubule (DCT)

- Actively transports NaCl (5-10% via a sodium-chloride co-transporter)
- Impermeable to water
 - ✓ The tubular fluid in the DCT is hypotonic regardless of hydration status
- Thick ascending limb and the DCT are collectively called the *diluting segment of the nephron*
 - ✓ Both of them are impermeable to water.

Collecting duct

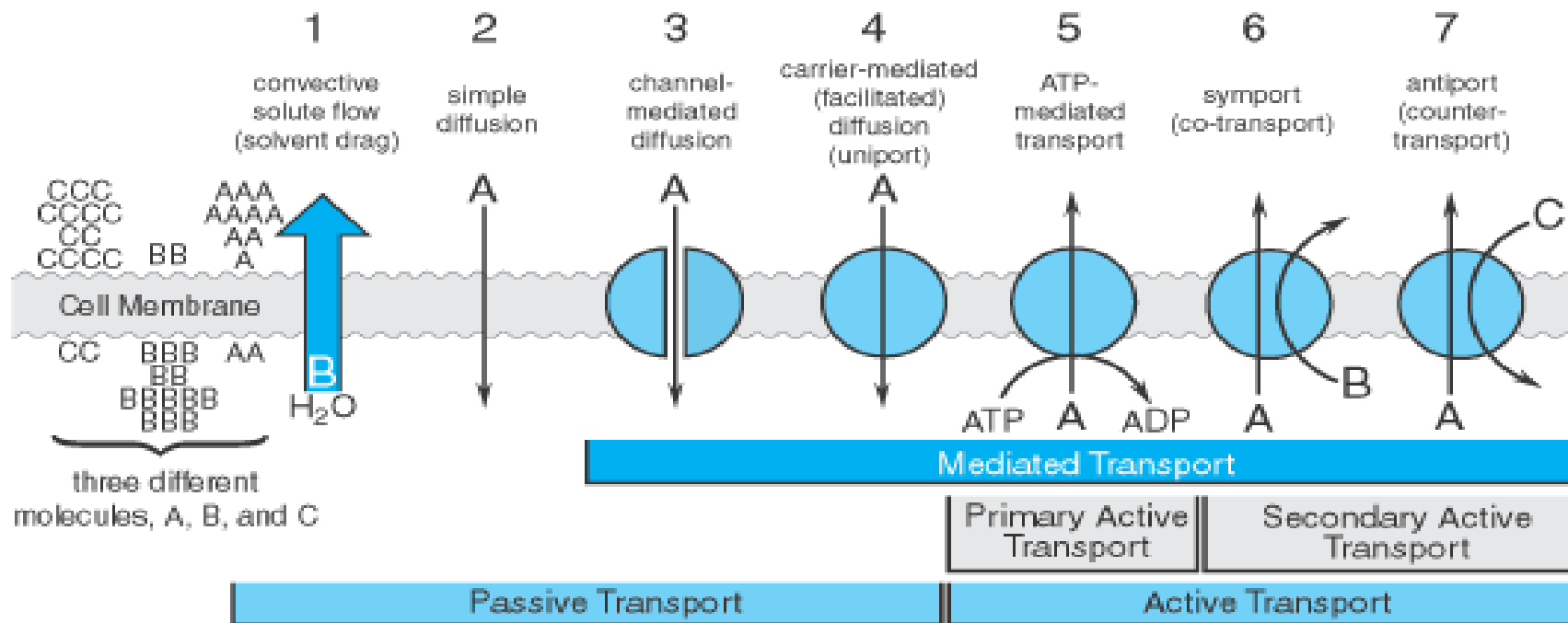
- An area of final control of ultrafiltrate composition and volume
- Final adjustments in electrolyte composition are made by 2 hormones
 - ✓ Adrenal steroid aldosterone
 - ✓ Antidiuretic hormone (ADH)
- Finally, less than **1%** of originally filtered sodium remains in the final urine.

3. Tubular secretion

- Involves in the elimination of organic chemicals
 - ✓ Organic Acid and
 - ✓ Organic Base
- Proximal tubule has a highly efficient transport system for organic acids and organic bases.
 - Powered by the sodium pump in the basolateral membrane

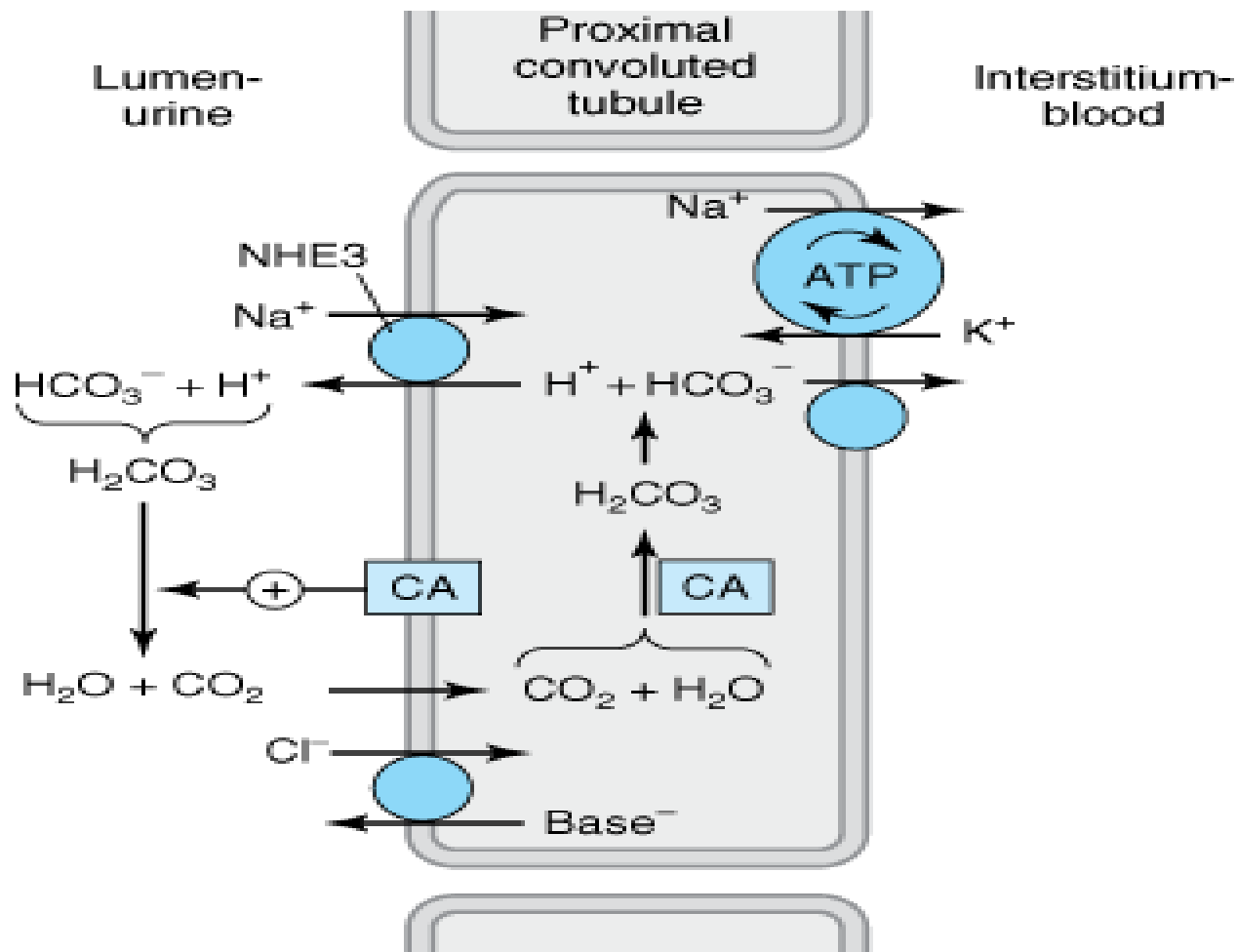
General Mechanism of Renal Epithelial Transport

- Seven mechanisms by which solutes may cross renal epithelial cell membranes



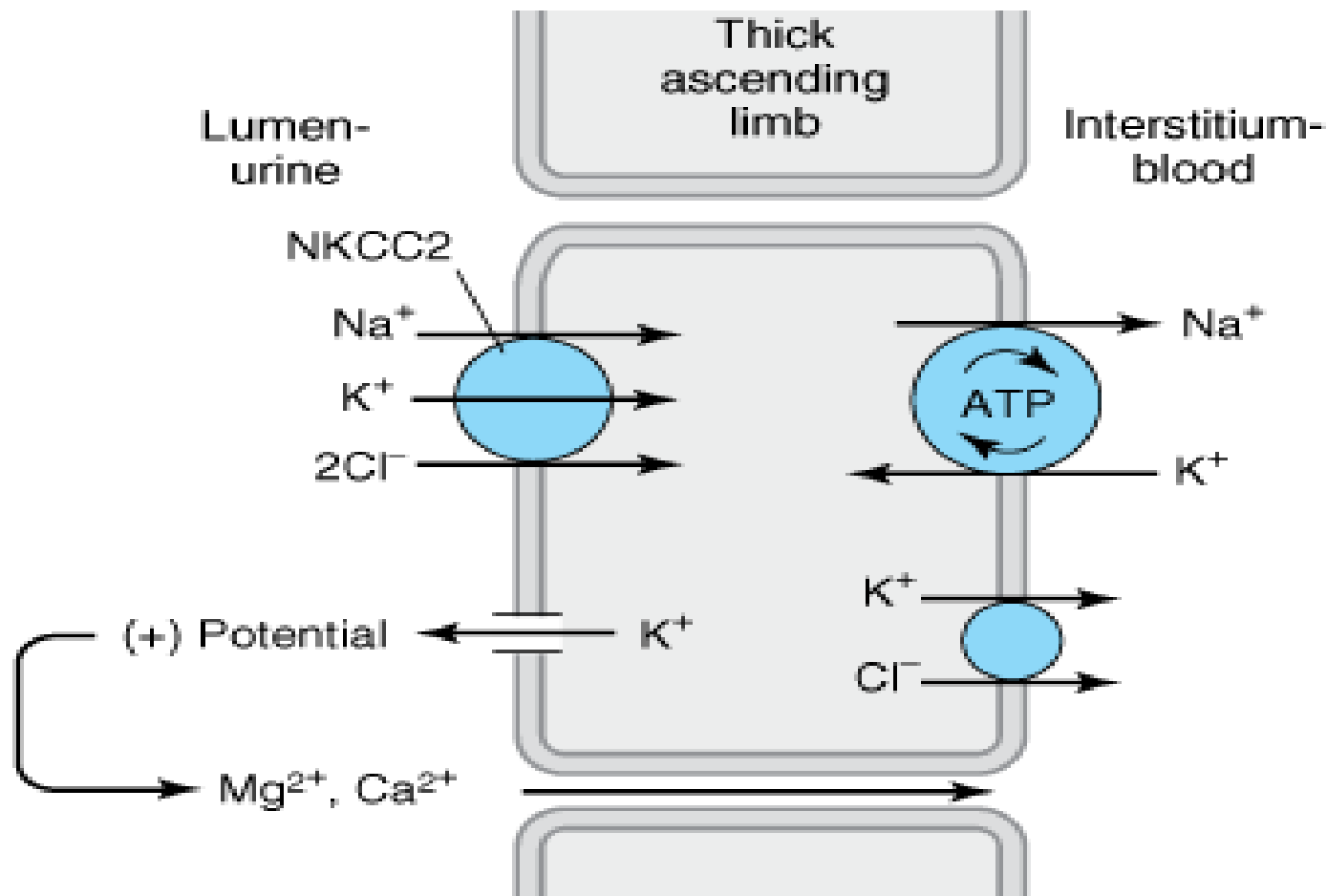
Mechanism of NaCl and NaHCO₃ reabsorption at the PCT

- 65% of total Na⁺, 85% of the filtered NaHCO₃, 65% of the K⁺, 60% of the water reabsorb at PCT
- Virtually all of the filtered glucose and amino acids are reabsorbed in the proximal tubule.
- What is the transport mechanism for NaHCO₃ and NaCl ???



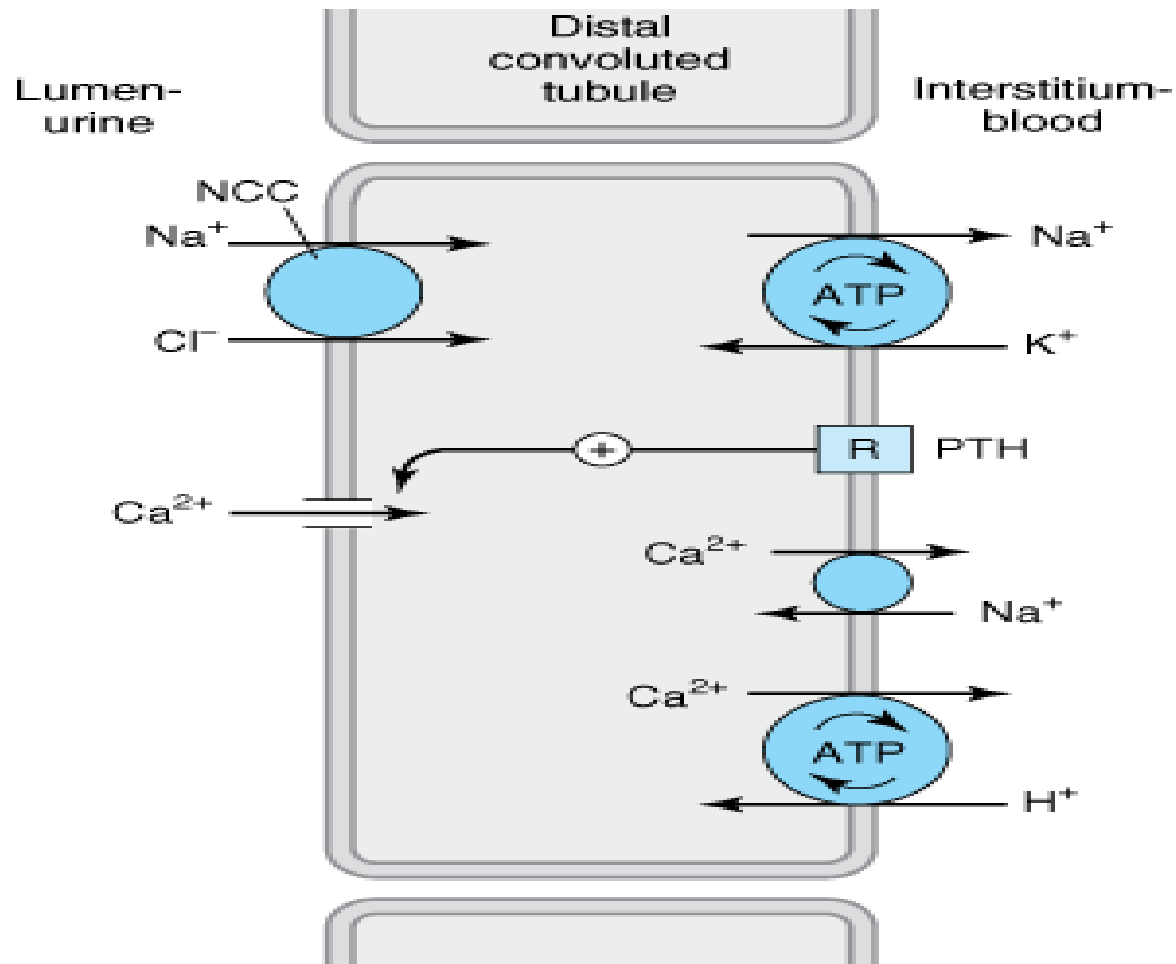
Mechanism of NaCl reabsorption at the TAL

- **25% of filtered Na^+** is reabsorbed mostly in the TAL
- The NaCl transport system in the luminal membrane of the TAL is a $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ cotransporter (called NKCC2 or NK2CL)
 - ✓ Site of second major luminal transport of NaCl
 - ✓ But contributes to excess K^+ accumulation within the cell
- The increased intracellular K^+ and Cl^- will exit basolateral membrane by K^+/Cl^- cotransporter.



Mechanism of NaCl reabsorption at DCT

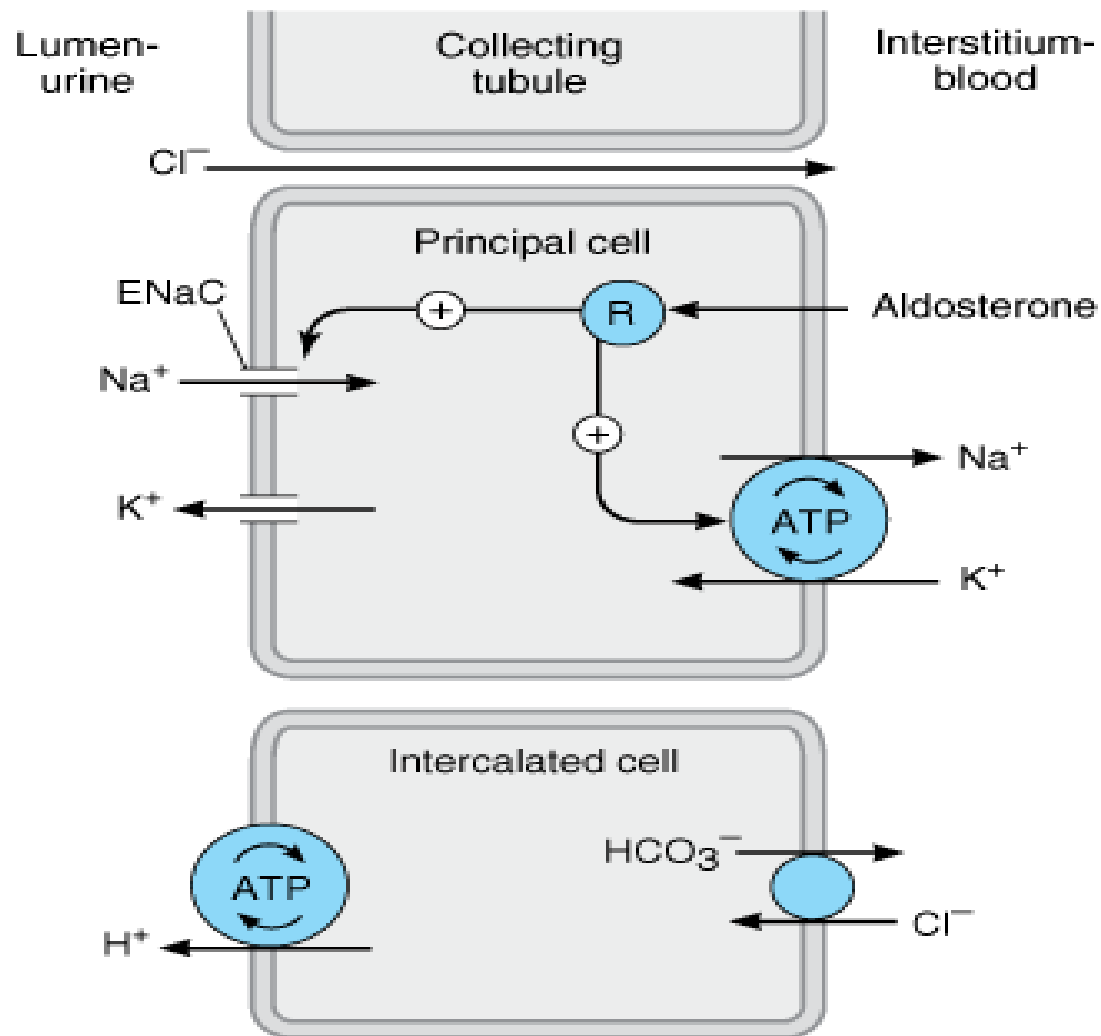
- 10% of the filtered NaCl reabsorption
- The mechanism of NaCl transport in the DCT is by an electrically neutral **Na⁺ and Cl⁻ cotransporter**
- Cl⁻ passively exits the basolateral membrane via Cl⁻ channel. (not shown in the figure)
- No lumen-positive potential in this segment
 - ✓ B/c K⁺ does not recycle across the apical membrane
- Ca²⁺ is actively reabsorbed via an apical Ca²⁺ channel and basolateral Na⁺/Ca²⁺ exchanger



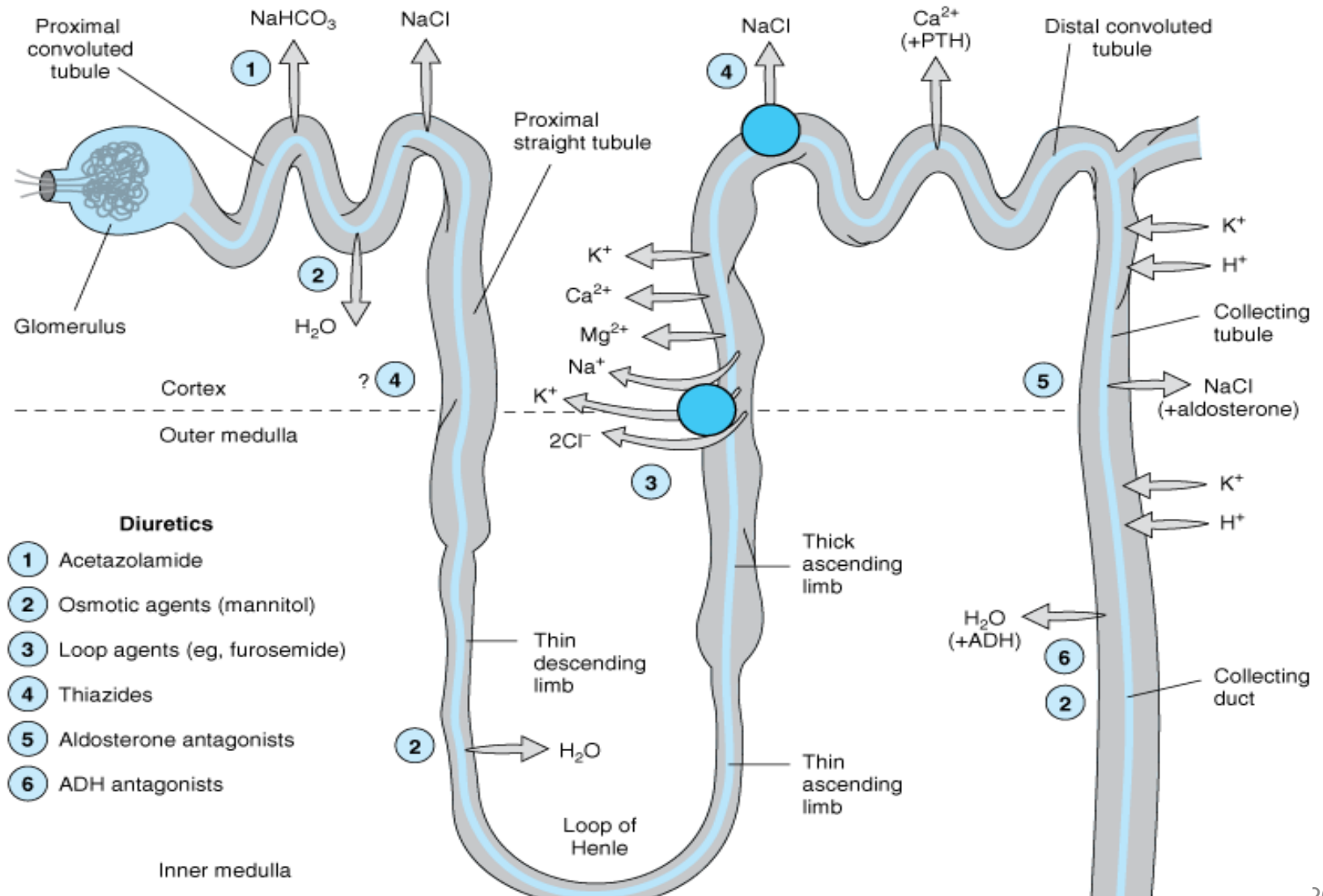
Mechanism of NaCl reabsorption at collecting tubule (CCT)

- CCT
 - ✓ Responsible for **only 2–5% of NaCl reabsorption**
 - ✓ Final site of NaCl reabsorption
 - ✓ Tight regulation of body fluid volume and for determining the final Na^+ concentration of the urine
 - ✓ Site at **which mineralocorticoids exert a significant influence**
 - ✓ **Most important site of K^+ secretion**
- Two types of cell at CCT
 - ❖ **Principal cells** are the major sites of Na^+ , K^+ , and water transport
 - ❖ **Intercalated cells** are the primary sites of H^+ secretion

- Because Na^+ entry into the principal cell predominates over K^+ secretion, a 10–50 mV lumen-negative electrical potential develops.
- This lumen-negative electrical potential
 - ✓ Drives the transport of Cl^- back to the blood via the paracellular pathway
 - ✓ Draws K^+ out of cells through K^+ channel.
 - ✓ Also aids the secretion of H^+ into lumen from intercalated cell.



Summary

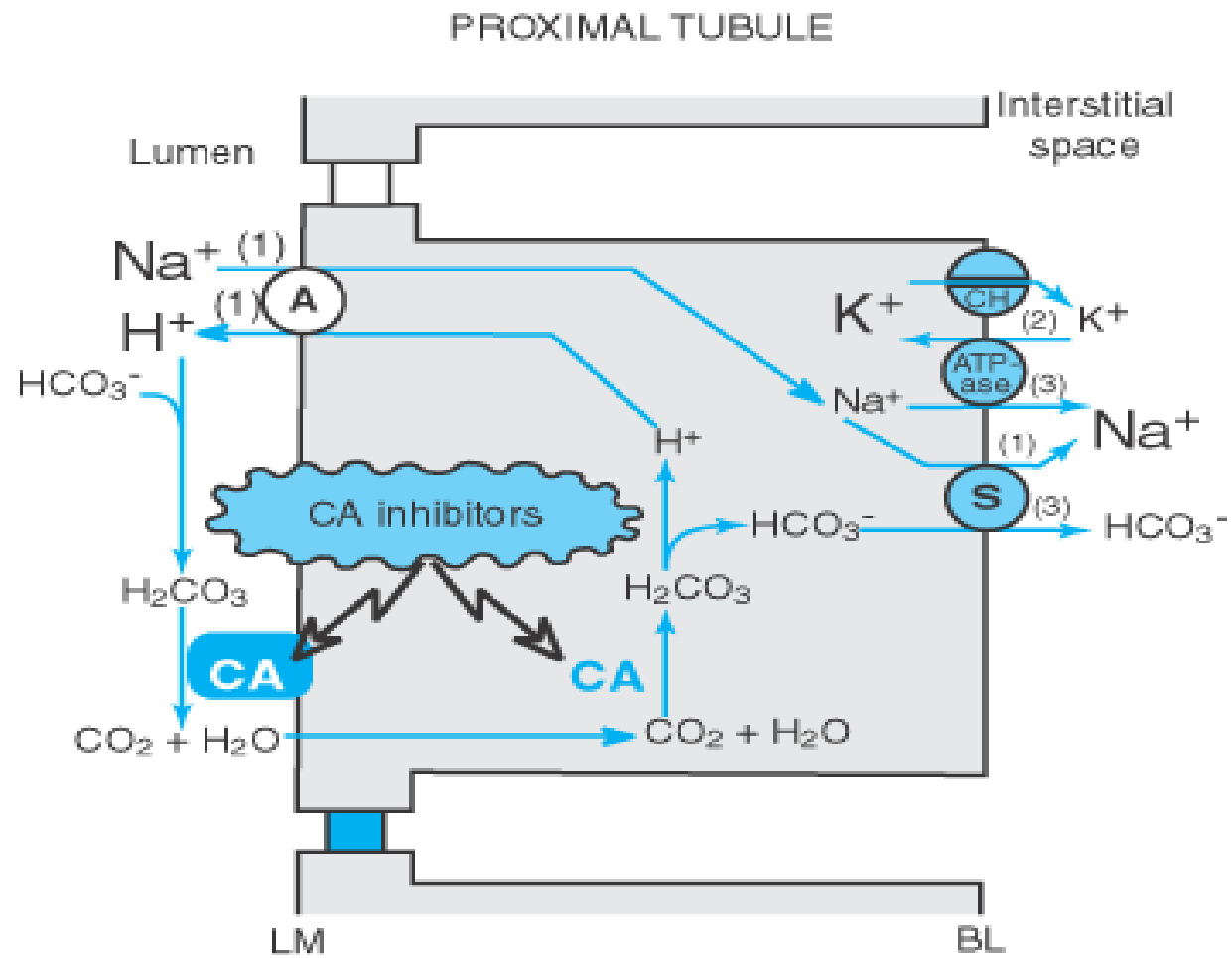


PRINCIPLES AND CLASSIFICATION OF DIURETICS

- By definition, diuretics are drugs that increase the rate of urine flow and sodium excretion
 - ✓ By altering how the kidney handles sodium (natriuresis)
- Combination of two diuretics (synergistic effect).
 - ✓ One nephron segment can compensate for altered Na^+ reabsorption at another nephron segment
 - ✓ Therefore, blocking multiple nephron sites significantly enhances efficacy.
- Classification is based on the **mechanism** and **site** of action.
 - ✓ **5 classes**

1. Carbonic Anhydrase Inhibitors

- Acetazolamide is the prototype of a class of agents.
- Other include dichlorophenamide , dorzolamide, brinzolamide and methazolamide
- ***Mechanism of diuretic action and effect on electrolyte excretion***
- Potently inhibit both the membrane-bound and cytoplasmic CA
 - ✓ indirectly inhibit luminal Na^+/H^+ pump
- Have little or no effect on the excretion of Ca^{2+} or Mg^{2+} .



Diuretic effect of the CA inhibitors is mild. WHY?

- ✓ Partially increased NaCl reabsorption in the later segments of the tubule (compensatory mechanism)
- ✓ Some HCO_3^- can still be absorbed at other nephron sites by *carbonic anhydrase-independent* mechanisms (65% of HCO_3^- is rescued)
- ✓ High percentage of enzyme activity must be inhibited
- ✓ Uncatalysed reaction between CO_2 and H_2O is sufficient to achieve HCO_3^- reabsorption.

Toxicity, Adverse Effects

➤ Hypokalemia

- Counteracted by simultaneous administration of KCl

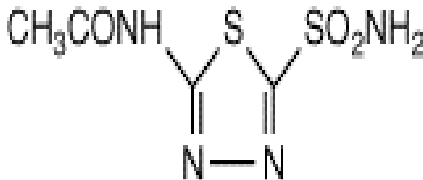
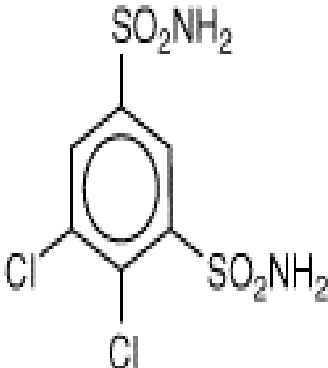
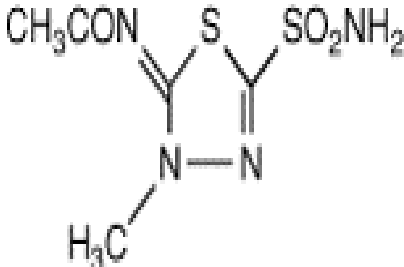
➤ Metabolic acidosis

- Develops due to **urinary loss of bicarbonate.**
- Responsible for most adverse effects, contraindications, and drug interactions
 - ✓ Hepatic encephalopathy (diversion of ammonia)
 - ✓ Renal Stones
 - Ca^{2+} salts are relatively insoluble at alkaline Ph
 - ✓ Respiratory acidosis

Therapeutic Uses

- Glaucoma (dorzolamide)
 - CA in the eye mediates the formation of large amounts of HCO_3^- in aqueous humor.
 - Inhibition of CA decreases the rate of formation of aqueous humor and consequently reduces intraocular pressure (IOP).
- Metabolic alkalosis
- Acute mountain sickness (AMS)
 - ✓ Increased bicarbonate excretion leads to *decreased blood pH*
 - ✓ Decreased serum pH lowers hemoglobin's affinity for oxygen

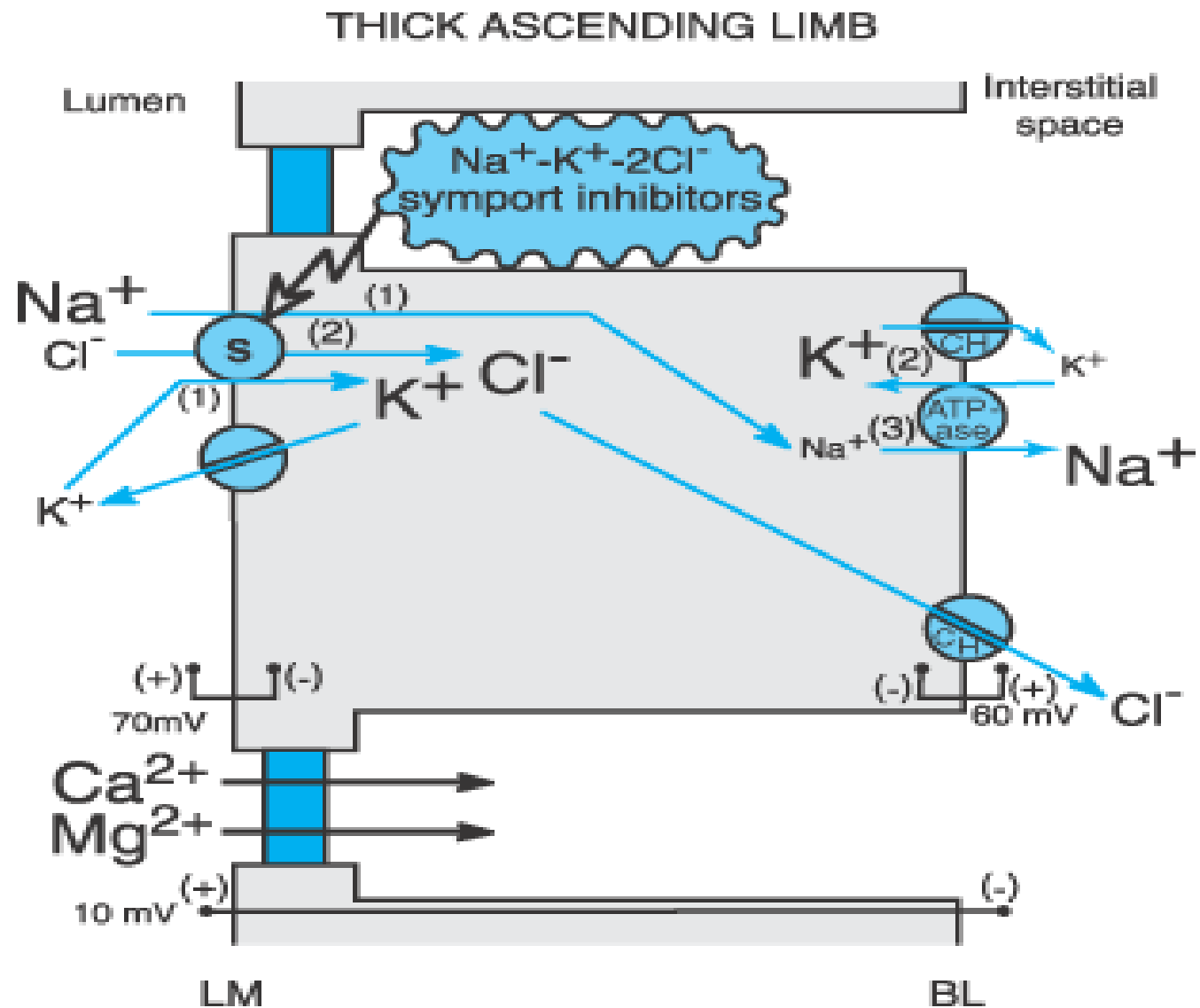
Pharmacokinetics

DRUG	STRUCTURE	RELATIVE POTENCY	ORAL AVAILABILITY	$T_{1/2}$ (HOURS)	ROUTE OF ELIMINATION
Acetazolamide (DIAMOX)		1	~100%	6-9	R
Dichlorophenamide (DARAMIDE)		30	ID	ID	ID
Methazolamide (GLAUCTABS)		>1; <10	~100%	~14	~25%, ~75% M

Abbreviations: R, renal excretion of intact drug; M, metabolism; ID, insufficient data.

2. Loop Diuretics, High-Ceiling Diuretics

- Typical loop diuretics is Furosemide
- Other agents include: bumetanide, ethacrynic acid and torsemide
- ***Mechanism of diuretic action***
 - Inhibit the activity of the $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ symporter in the TAL
 - ✓ Interfere with the diluting and concentrating mechanisms of the kidney.
 - Also inhibit Ca^{2+} and Mg^{2+} reabsorption, HOW?
 - ✓ Abolish the transepithelial potential difference
 - Increase the urinary excretion of K^+ and titratable acid.
 - ✓ Due in part to increased delivery of Na^+ to the distal tubule



mechanism cont'd...

- Are also called *high-ceiling diuretics*.

Because

- ✓ TAL has **a great reabsorptive capacity** and reabsorbs most of the rejectate from the proximal tubule.
- ✓ Nephron segments past the TAL do not possess sufficient reabsorptive capacity

Toxicity, Adverse Effects, Drug Interactions

- Hyponatremia (serious depletion of total-body Na^+)
- Hypokalemia and metabolic alkalosis
 - ✓ May induce cardiac arrhythmias
- Hypomagnesemia (a risk factor for cardiac arrhythmias) and rare hypocalcemia
 - ✓ Avoided in postmenopausal osteopenic women
- Ototoxicity: dose-related hearing loss characterized by hearing impairment and deafness. (common with Ethacrynic acid)
- Hyperuricemia (occasionally leading to gout)

Drug interactions

- **NSAIDs** attenuate the diuretic response to loop diuretics
 - ✓ In part by preventing prostaglandin-mediated increases in RBF
- **Aminoglycosides** (synergism of ototoxicity caused by both drugs),
- Digitalis glycosides (increased digitalis-induced arrhythmias),
- cisplatin (increased risk of diuretic-induced ototoxicity),
- Probenecid (blunted diuretic response),
- Thiazide diuretics (synergism of diuretic activity of both drugs leading to profound diuresis), and
- Amphotericin B (increased potential for nephrotoxicity and toxicity and intensification of electrolyte imbalance).

Therapeutic Uses

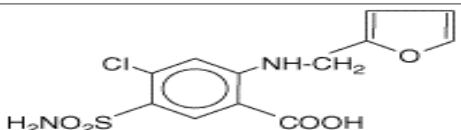
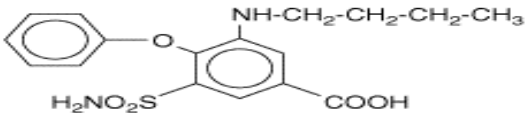
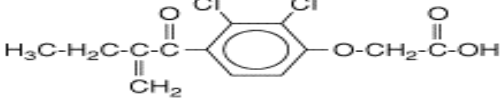
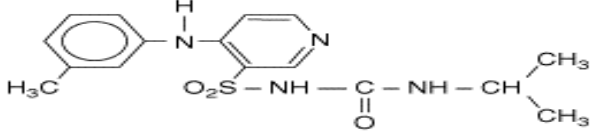
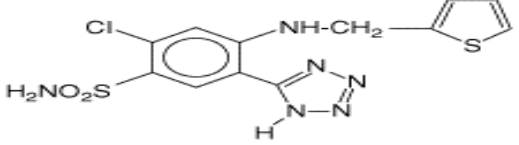
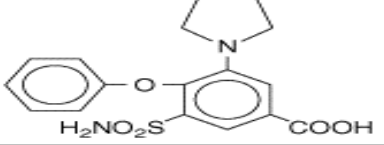
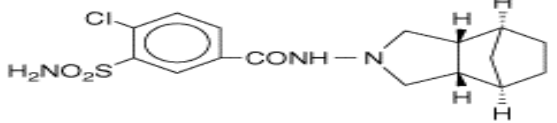
- Treatment of edematous condition
 - ✓ Acute pulmonary edema
 - Natriuresis reduces left ventricular filling pressures
 - ✓ Chronic congestive heart failure
 - By reducing extracellular fluid volume
- Treatment of hypertension
 - ✓ Used when patients do not respond satisfactorily to thiazide diuretics and anti-hypertensive drugs.
 - B/c of short elimination half-lives
- Hyperkalemia

P`kinetics

- All of the loop diuretics are available for both oral and parenteral administration.
- Their onset of action is rapid, usually within **30 minutes after oral and 5 minutes after intravenous administration.**
- They produce peak diuresis in about 2 hours,
- Total duration of diuretic action of approximately 6 to 8 hours.
- They are extensively bound to plasma proteins
- They are eliminated in the urine by both glomerular filtration and tubular secretion.
- Only small amounts of these compounds appear to be metabolized by the liver.

Pharmacokinetics

Table 28-4. Inhibitors of $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ Symport (Loop Diuretics, High-Ceiling Diuretics)

DRUG	STRUCTURE	RELATIVE POTENCY	ORAL AVAILABILITY	$T_{1/2}$ (HOURS)	ROUTE OF ELIMINATION
Furosemide (LASIX)		1	~60%	~1.5	~65% R, ~35% M*
Bumetanide (BUMEX)		40	~80%	~0.8	~62% R, ~38% M
Ethacrynic acid (EDECRIN)		0.7	~100%	~1	~67% R, ~33% M
Torsemide (DEMADEX)		3	~80%	~3.5	~20% R, ~80% M
Axosemide*		1	~12%	~2.5	~27% R, 63% M
Piretanide*		3	~80%	0.6-1.5	~50% R, ~50% M
Tripamide*		ID	ID	ID	ID

*Not available in the United States.

*For furosemide, metabolism occurs predominantly in the kidney.

Abbreviations: R, renal excretion of intact drug; M, metabolism; ID, insufficient data.

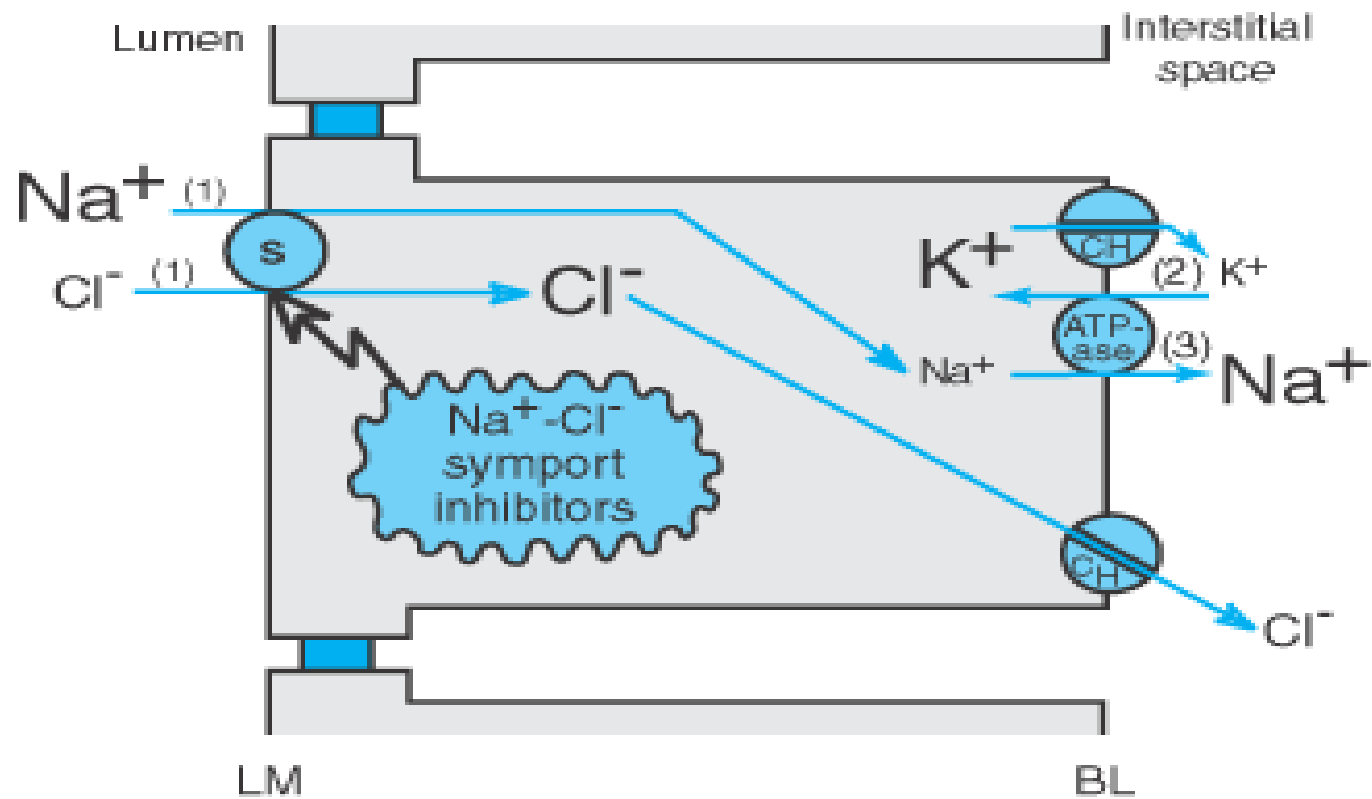
3. Thiazide and thiazide like diuretics

- The prototype thiazide is hydrochlorothiazide
- Other include: bendroflumethiazide, benzthiazide, chlorothiazide, hydrochlorothiazide, hydroflumethiazide, methyclothiazide, polythiazide, and trichlormethiazide
- **Thiazidlike compounds**; chlorthalidone, Indapamide, quinethazone, and metolazone.

➤ ***Mechanism of diuretic action***

- Bind and inhibit the $\text{Na}^+\text{-Cl}^-$ symporter at the DCT
- Have only a medium diuretic efficacy.
 - ✓ Maximum excretion of filtered load of Na^+ is only 5%
 - ✓ Large amount (up to 90%) is absorbed at the earlier sites.
- Increase the excretion of K^+ and titratable acid
- Increase Ca^{2+} reabsorption, cause a mild magnesuria

DISTAL CONVOLUTED TUBULE

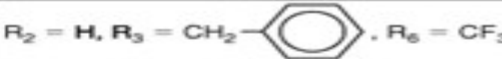
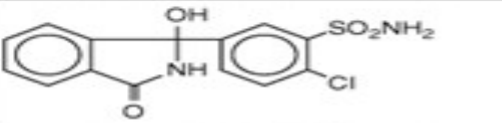
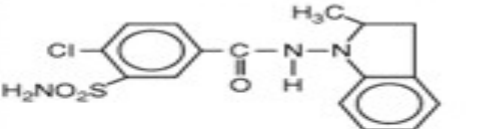
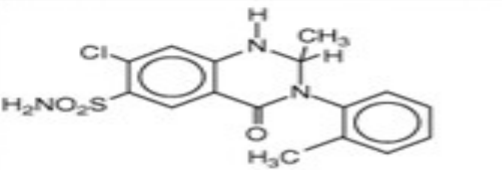
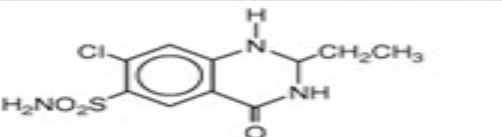


- At higher doses, administration of some of the thiazides results in some degree of carbonic anhydrase inhibition.
- However, *at usual doses, only chlorothiazide shows any appreciable carbonic anhydrase inhibitory activity.*

- Orally administered thiazides are rapidly absorbed from the GIT and begin to produce diuresis in about 1 hour.
- Approximately 50% of an oral dose is excreted in the urine within 6 hours.
- These compounds are organic acids and are actively secreted into the proximal tubular fluid by the organic acid secretory mechanism.
- There also appears to be an extrarenal pathway for their elimination involving the hepatic–biliary acid secretory system that is particularly important for thiazide elimination when renal function is impaired.

P'cokinetics

Inhibitors of Na⁺-K⁺ Symport (Thiazide and Thiazidelike Diuretics)

DRUG	STRUCTURE	RELATIVE POTENCY	ORAL AVAILABILITY	t _{1/2} (HOURS)	ROUTE OF ELIMINATION
Bendroflumethiazide (NATURETIN)	 $R_2 = H, R_3 = CH_2 - \text{C}_6\text{H}_5, R_6 = CF_3$	10	~100%	3-3.9	~30% R, ~70% M
Chlorothiazide (DIURIL)	$R_2 = H, R_3 = H, R_6 = Cl$ (Unsaturated between C3 and N4)	0.1	9-56% (dose-dependent)	~1.5	R
Hydrochlorothiazide (HYDRODIURIL)	$R_2 = H, R_3 = H, R_6 = Cl$	1	~70%	~2.5	R
Hydroflumethiazide (SALURON)	$R_2 = H, R_3 = H, R_6 = CF_3$	1	~50%	~17	40-80% R, 20-60% M
Methyclothiazide (ENDURON)	$R_2 = CH_3, R_3 = CH_2Cl, R_6 = Cl$	10	ID	ID	M
Polythiazide (RENESE)	$R_2 = CH_3, R_3 = CH_2SCH_2CF_3, R_6 = Cl$	25	~100%	~25	~25% R, ~75% U
Trichlormethiazide (NAQUA)	$R_2 = H, R_3 = CHCl_2, R_6 = Cl$	25	ID	2.3-7.3	R
Chlorthalidone (HYGROTON)		1	~65%	~47	~65% R, ~10% B, ~25% U
Indapamide (LOZOL)		20	~93%	~14	M
Metolazone (MYKROX, ZAROXOLYN)		10	~65%	ID	~80% R, ~10% B, ~10% M
Quinethazone (HYDROMOX)		1	ID	ID	ID

Abbreviations: R, renal excretion of intact drug; M, metabolism; B, excretion of intact drug into bile; U, unknown pathway of elimination; ID, insufficient data

Toxicity, Adverse Effects

- Abnormalities of fluid and electrolyte balance
 - Include extracellular volume depletion, hypotension, hypokalemia, hyponatremia, hypochloremia, metabolic alkalosis, hypomagnesemia, hypercalcemia, Hyperuricemia
- CNS disturbance (*e.g.*, vertigo, headache, paresthesias, xanthopsia, and weakness)
- Erectile dysfunction.
- Gastrointestinal disturbance, Hyperglycemia

- Treatment of mild to moderate hypertension
 - Either alone or in combination with other antihypertensive drugs, inexpensive, efficacious , administered once daily,
 - Are the best **initial therapy**
- Treatment of the edema
- Treatment of osteoporosis
- The mainstay for treatment of **nephrogenic diabetes insipidus (paradoxical effect)**
- Treatment of the edema associated with heart (congestive heart failure), and renal (nephrotic syndrome, chronic renal failure)
- Hepatic cirrhosis
 - *The thiazides remain the drugs of first choice*

- Nephrotic syndrome
 - proteinuria and
 - edema due to some form of glomerulonephritis.
 - Fall in plasma protein concentration decreases vascular volume, which leads to diminished renal blood flow.
- This in turn causes **secondary aldosteronism** characterized by **Na and water retention and K depletion**.
 - *Thiazide (possibly with a K- sparing diuretic) to control the secondary aldosteronism, is a useful initial approach to treatment.*
- Since nephrotic edema is frequently **more difficult to control than cardiac edema**,
 - it may be necessary to switch to a loop diuretic (and spironolactone) to obtain adequate diuresis

- *Premenstrual Edema and Edema of Pregnancy*
 - Excessive premenstrual edema frequently responds well to thiazide therapy.
 - *Recent experience has diminished enthusiasm for use of any diuretics in pregnant women.*

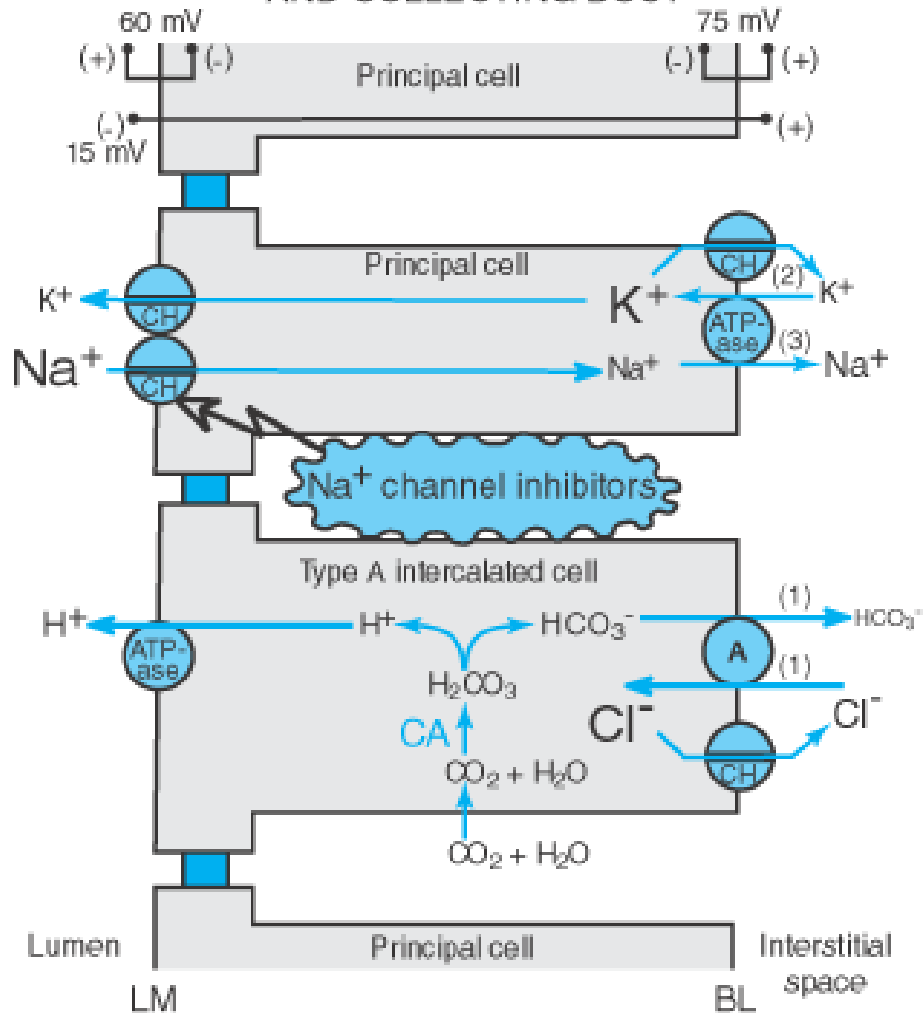
4. K⁺-Sparing Diuretics

- Two classes
 - Na⁺ Channels Inhibitors : Triamterene and amiloride
 - Aldosterone antagonist : Spironolactones, eplerenone, canrenone

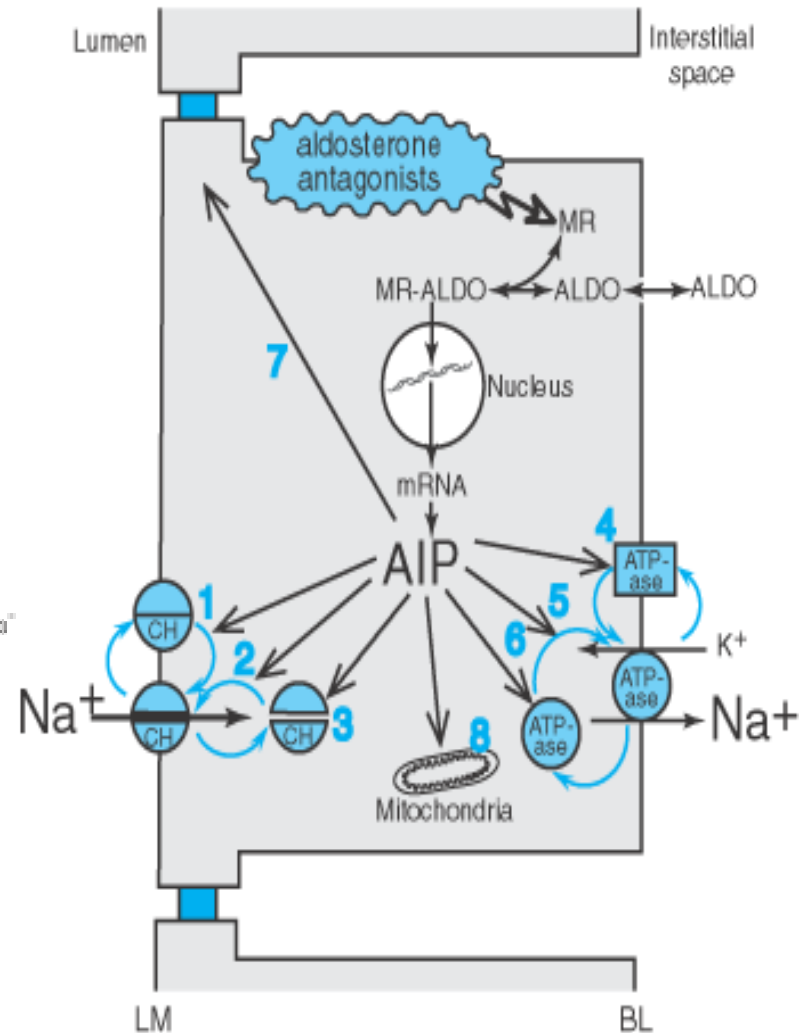
❖ *Mechanism of diuretic action*

- Triamterene and amiloride blocks epithelial Na⁺ channels (competing with Na⁺)
- Spironolactone and eplerenone competitively inhibit the binding of aldosterone to the MR
 - ✓ The only diuretics that do not require access to the tubular lumen to induce diuresis.

LATE DISTAL TUBULE AND COLLECTING DUCT



LATE DISTAL TUBULE AND COLLECTING DUCT

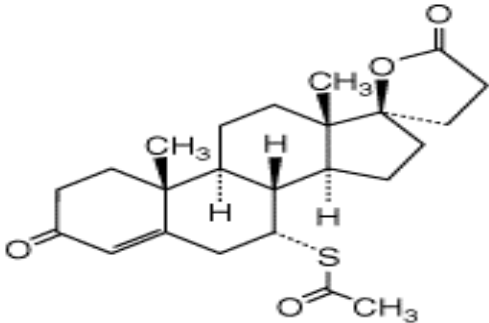
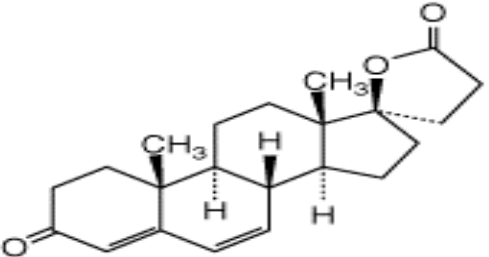
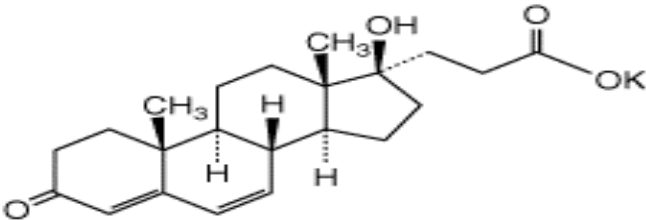
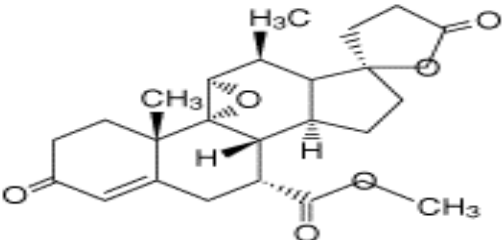


Mechan cont'd...

- Both class decreases the excretion rates of K^+ , H^+ , Ca^{2+} , and Mg^{2+} .
 - ✓ By attenuating the lumen-negative voltage
- Limited efficacy
 - ✓ Limited capacity to reabsorb solutes (approximately 2% of filtered load).
- However, the diuretic activity increased if :
 - ✓ Aldosterone concentration are high
 - ✓ Sodium loads in the body is high
 - ✓ Sodium load in the tubule is high secondary to diuresis

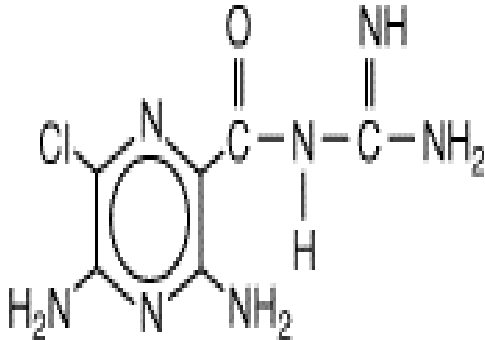
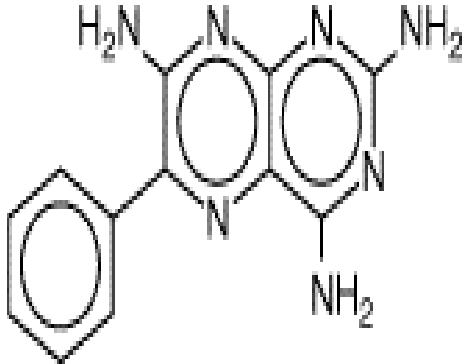
Pharmacokinetics

Mineralocorticoid Receptor Antagonists (Aldosterone Antagonists, Potassium-Sparing Diuretics)

DRUG	STRUCTURE	ORAL AVAILABILITY	$t_{1/2}$ (HOURS)	ROUTE OF ELIMINATION
Spironolactone (ALDACTONE)		~65%	~1.6	M
Canrenone*		ID	~16.5	M
Potassium canrenoate*		ID	ID	M
Eplerenone (INSPIRA)		ID	~5	M

*Not available in United States

- Na⁺ Channels Inhibitors

DRUG	STRUCTURE	RELATIVE POTENCY	ORAL AVAILABILITY	<i>t</i> 1/2 (HOURS)	ROUTE OF ELIMINATION
Amiloride (DYRENIUM)		1	15-25%	~21	R
Triamterene (MIDAMOR)		0.1	~50%	~4.2	M

Abbreviations: R, renal excretion of intact drug; M, metabolism; however, triamterene is transformed into an active metabolite that is excreted in the urine

Therapeutic Uses

- Treatment of edema or hypertension
 - Not alone but in combination with other diuretics.
(B/c of mild natriuresis)
 - Combination with thiazide or loop diuretic
 - ✓ Mainly to avoid hypokalemia and to enhance diuretic efficacy.
- Treatment hyperaldosteronism
 - Spironolactone is particularly useful in the treatment of **primary hyperaldosteronism** and **secondary hyperaldosteronism**
 - Spironolactone -diuretic of choice in patients with hepatic cirrhosis

Toxicity, Adverse Effects, Contraindications

- Hyperkalemia
 - Most dangerous and can be life-threatening.
 - Contraindicated in patients with hyperkalemia, patients taking K^+ supplements, renal failure...
- Metabolic acidosis which occur due to decrease secretion of H^+
- CNS, gastrointestinal, musculoskeletal, dermatological,
- Spironolactone may cause gynecomastia, impotence, decreased libido, hirsutism, deepening of the voice, and menstrual irregularities.

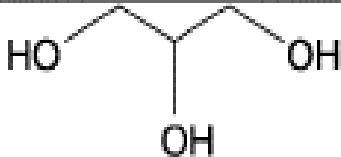
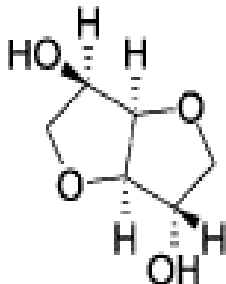
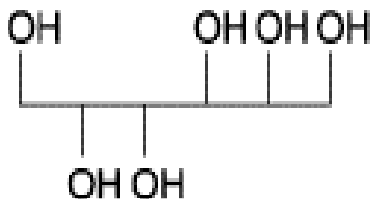
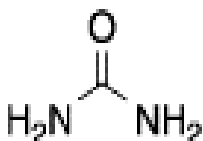
5. Osmotic Diuretics

- Agents that are freely filtered at the glomerulus, undergo limited reabsorption by the renal tubule, and are relatively inert pharmacologically.
- **Example: Polyols**
- The prototype osmotic agent is mannitol
- Other drugs include: glycerin, sorbitol, isosorbide (bicyclic and basic form of sorbitol) and urea.
- ***Mechanism of diuretic action***
- By increasing tubular tonicity, they extract water from intracellular compartments, resulting in expansion of the extracellular fluid volume.
 - ✓ Reduce medullary tonicity
- Limit the osmosis of water into the interstitial space

Mechanism cont'd...

- Act both in the proximal tubule and the **loop of Henle (DTL) (major site of action)**
- Increase the urinary excretion of nearly all electrolytes, including Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Cl^- , HCO_3^- and phosphate.

Osmotic Diuretics

DRUG	STRUCTURE	ORAL AVAILABILITY	$t_{1/2}$ (HOURS)	ROUTE OF ELIMINATION
Glycerin (OSMOGLYN)		Orally active	0.5-0.75	~80% M
				~20% U
Isosorbide (ISMOTIC)		Orally active	5-9.5	R
Mannitol (OSMITROL)		Negligible	0.25-1.7*	~80% R
				~20% M + B
Urea (UREAPHIL)		Negligible	ID	R

*In renal failure, 6-36.

Abbreviations: R, renal excretion of intact drug; M, metabolism; B, excretion of intact drug into bile; U, unknown pathway of elimination; ID, insufficient data.

Toxicity, Adverse Effects

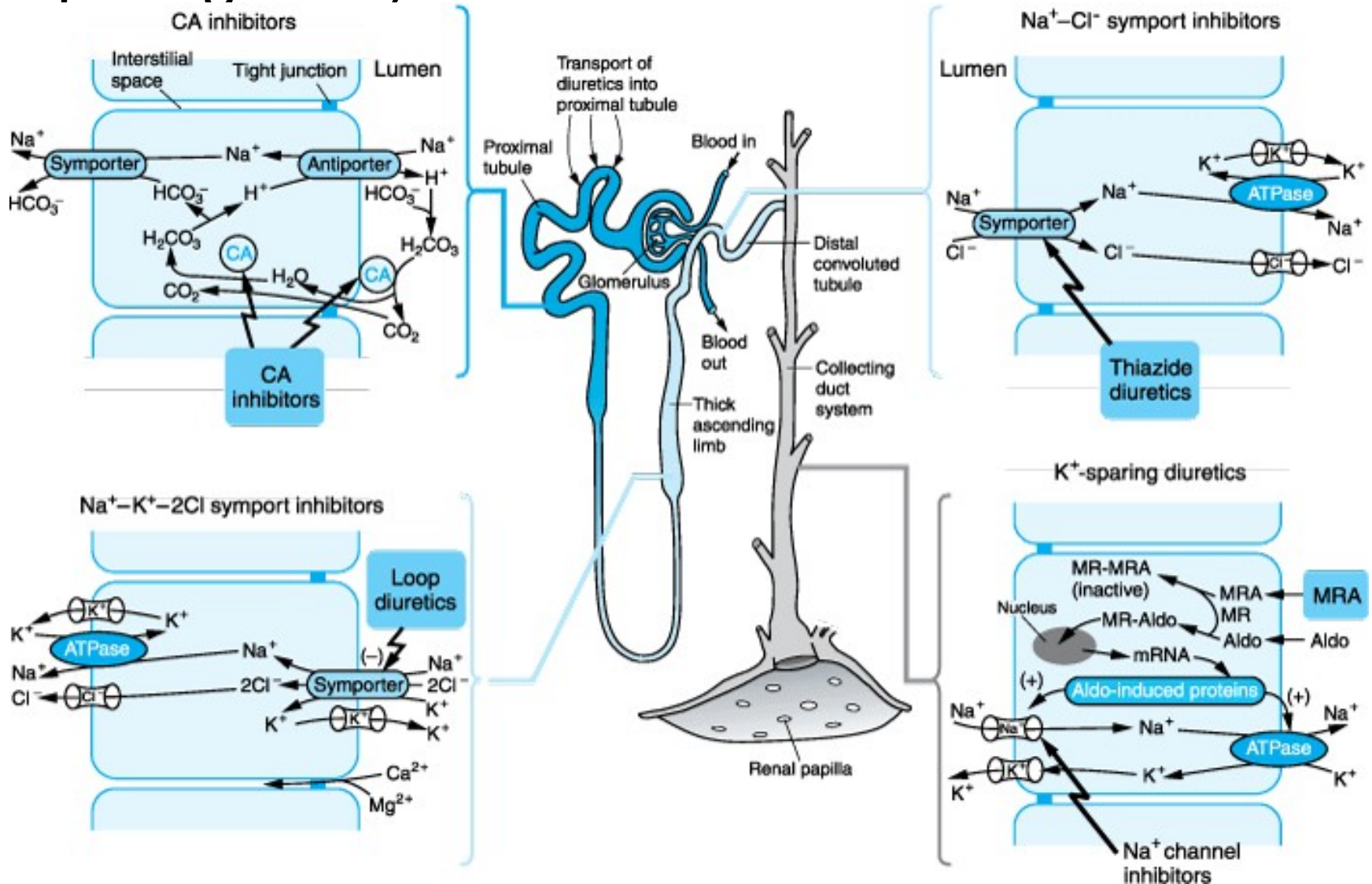
- Extracellular fluid volume expansion
 - Cause pulmonary edema and aggravate heart failure or pulmonary congestion.
- Hypernatremia and dehydration
 - Excess use may lead to free water loss, severe dehydration and hypernatremia
- Both mannitol and urea are contraindicated in patients with active cranial bleeding

- Acute renal failure (ARF)
 - ✓ Remove of obstructing tubular casts,
 - ✓ Dilution of tubular nephrotoxic substances
 - ✓ Reduction of swelling of tubular elements *via* osmotic extraction of water
 - ✓ to prevent complete renal shutdown.
 - ***IV mannitol*** generally being the agent of choice.
- Reduction of intracranial and IOP
 - ✓ Extract water from the eye and brain.
 - *The parenteral administration of a hypertonic solution of one of the osmotic diuretics.*
- Reduce cerebral edema

Resistance to Diuretic Administration

- Failure to respond higher doses of loop diuretics(80 mg of furosemide once or twice daily or greater in those with reduced GFR or HF)
- Combinations of diuretics may be used as an alternative approach to treating diuretic resistance
- Na restriction is being followed
- Administration of CAIs may be sufficient to enhance Na delivery to TAL, where its reabsorption can be blocked by loop diuretics.
- Thiazide diuretics may be combined with the loop diuretic to limit absorption by DCTs.
- The thiazide like diuretic **metolazone**, which has some proximal tubule effects unrelated to carbonic anhydrase, appears to be the most effective of the thiazide and Thiazide like drugs in this regard

Summary (loop>thiazide>K⁺ sparing>CAI)



1. When a patient is treated with a thiazide diuretic for hypertension, all of the following are likely EXCEPT:

- (A) The fall of blood pressure that occurs in the first 2 weeks of therapy results from a decrease of extracellular volume.
- (B) The sustained fall in blood pressure that occurs after several weeks of therapy is due to a decrease of intravascular resistance.
- (C) After the blood pressure is reduced, hypokalemia remains a complication.
- (D) Hyperuricemia may occur.
- (E) Hypoglycemia may occur.

2. Furosemide increases the excretion of all of the following EXCEPT:

- (A) Na^+
- (B) K^+
- (C) Ca^{2+} and Mg^{2+}
- (D) Uric acid

3. Which of the following drugs is an appropriate initial antihypertensive therapy in an otherwise healthy adult with mild hypertension?

- (A) Bumetanide
- (B) Triamterene
- (C) Hydrochlorothiazide
- (D) Aldacton (spironolactone)

4. When furosemide is administered to a patient with pulmonary edema, there is often symptomatic relief within 5 minutes of starting treatment. This relief is primarily due to:

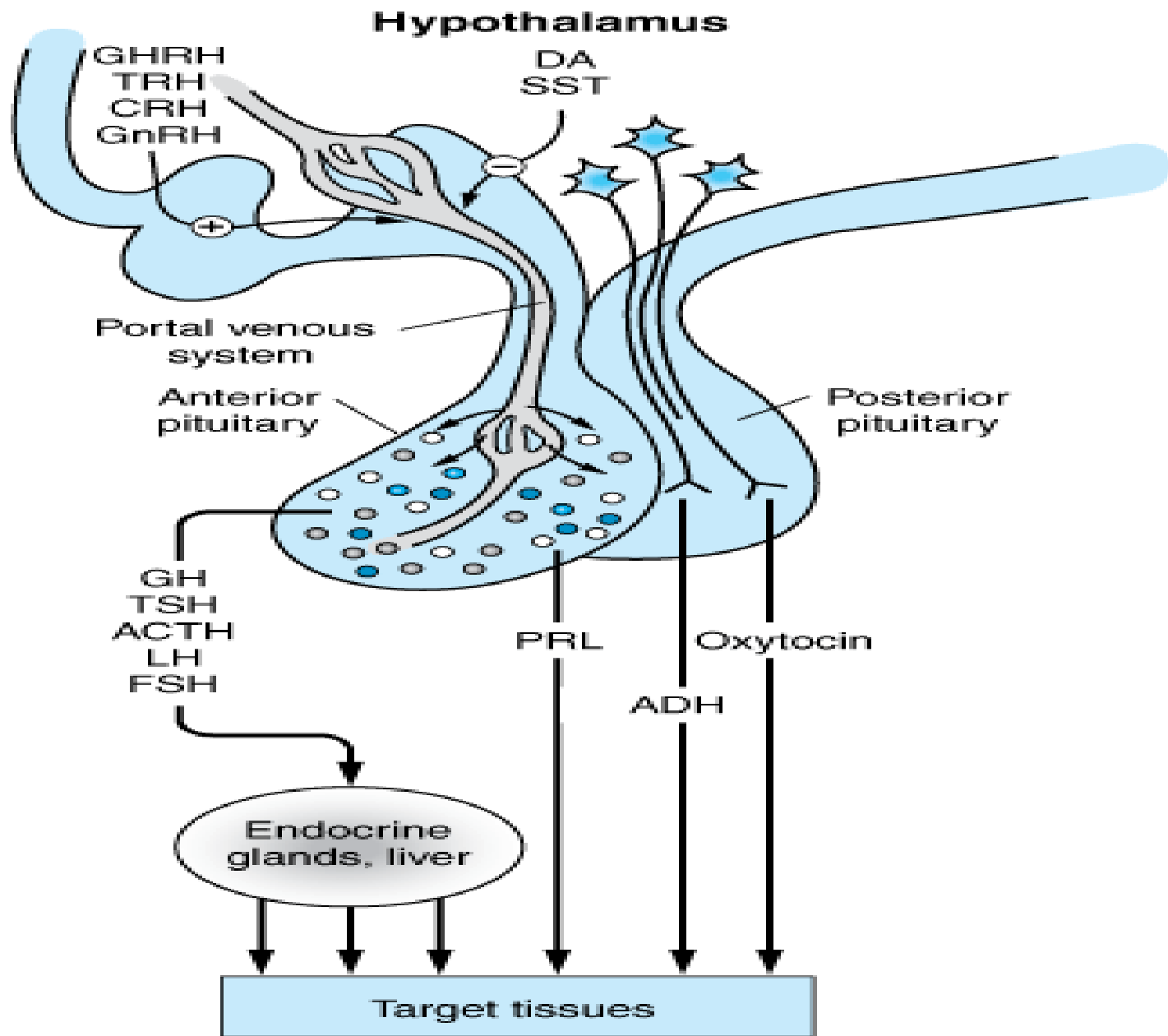
- (A) A rapid diuretic effect
- (B) An increase in venous capacitance
- (C) A direct effect on myocardial contractility
- (D) Psychological effects

5. All of the following statements are true regarding patients with renal insufficiency who exhibit a reduced diuretic response EXCEPT:

- (A) When the GFR drops below 30 mL/minute, thiazide diuretics are virtually useless.
- (B) The combination of a thiazide plus a potassium- sparing diuretic may yield an adequate diuretic response.
- (C) An 80-mg dose of IV furosemide followed an hour later by a 500-mg dose of IV chlorothiazide will probably yield the highest possible response.
- (D) Metolazone is contraindicated.

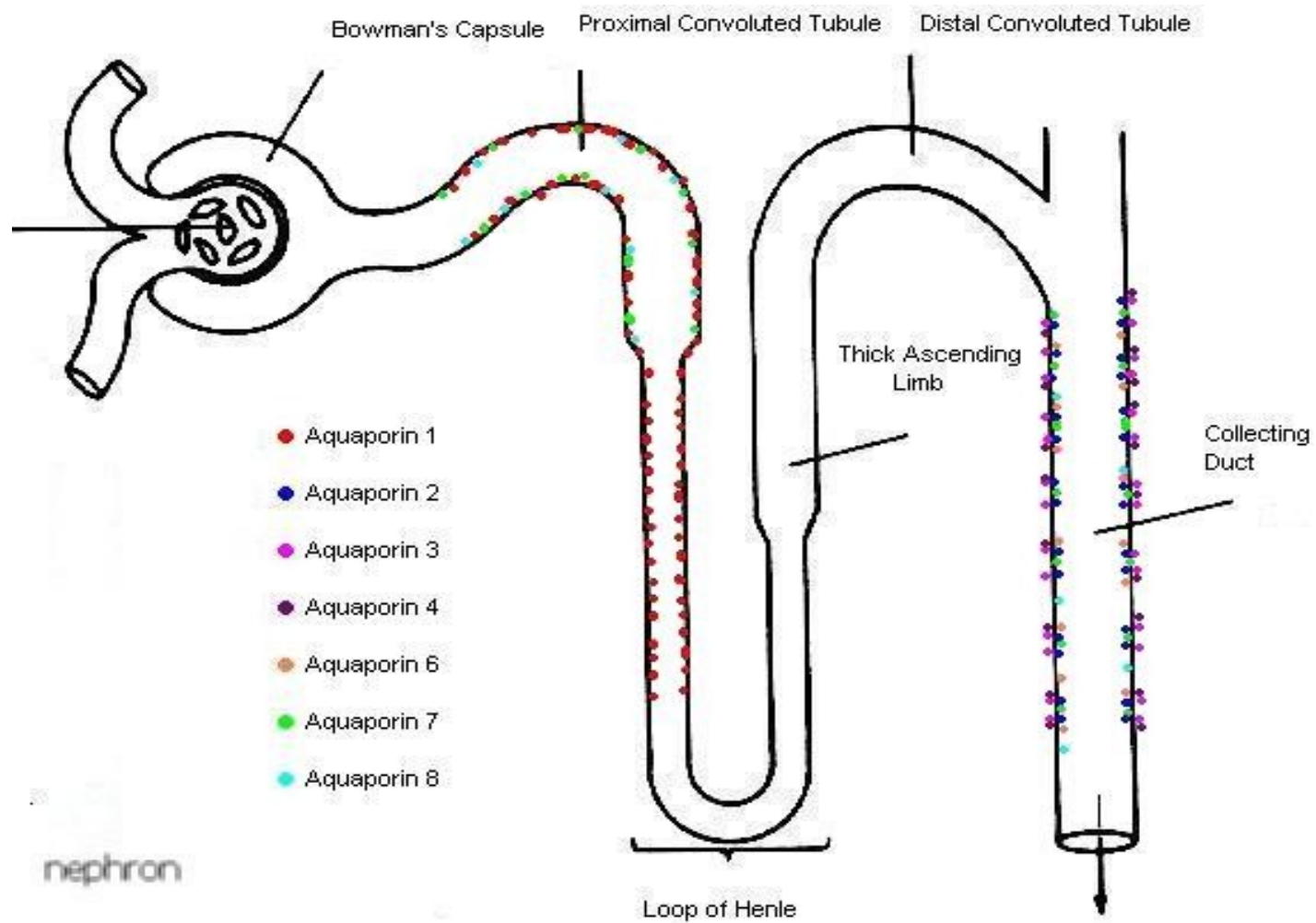
Vasopressin and Other Agents Affecting Renal Conservation of Water

- Arginine vasopressin (the antidiuretic hormone, ADH) is the main hormone that **regulates body fluid osmolality.**
- **Synthesized in the hypothalamus**
- Potent stimuli to its release
 - ✓ **Increased plasma osmolality** (principal physiological stimulus)
 - ✓ Sever hypovolaemia and/or hypotension.
 - ✓ Diminished activity of baroreceptor, carotic sinuses



3 subtypes

- V_1 (formerly known as V_{1a} receptors)
 - Are located **on vascular smooth muscle** and mediate **vasoconstriction**
- *V2 receptors*
 - Are located on the **basolateral membrane of distal tubule and collecting duct**,
 - Mobilize **aquaporin** channel, to be inserted into the apical membrane of the renal collecting duct cell.
- *V3 receptors* (formerly known as V_{1b} receptors)
 - Located in **pituitary gland and stimulate ACTH**



➤ Renal action

- **Promotes exocytosis** of water channel containing vesicles (aquaporin-2)
- More aqueous channel get inserted into the apical membrane.
- **Rate of endocytosis** and degradation of water channel containing vesicles is concurrently **reduced**
- Also **increase medullary hypertonicity** by stimulating vasopressin regulated urea transporter (VRUT) that increase urea permeability.

Action cont'd...

➤ Non-renal Actions

- **Vasopressin** is a potent **vasoconstrictor** (V_1 -receptor-mediated) though **V_2 receptors cause vasodilation in some blood vessels.**
 - ✓ Occur at concentration higher than those required for maximal antidiuresis
- Vasopressin plays a role as a neurotransmitter and/or neuromodulator
 - ✓ Mediated predominantly by V_1 receptors
- Cause platelet aggregation, contraction of uterine smooth muscle...

VASOPRESSIN RECEPTOR AGONISTS AND ANTAGONISTS

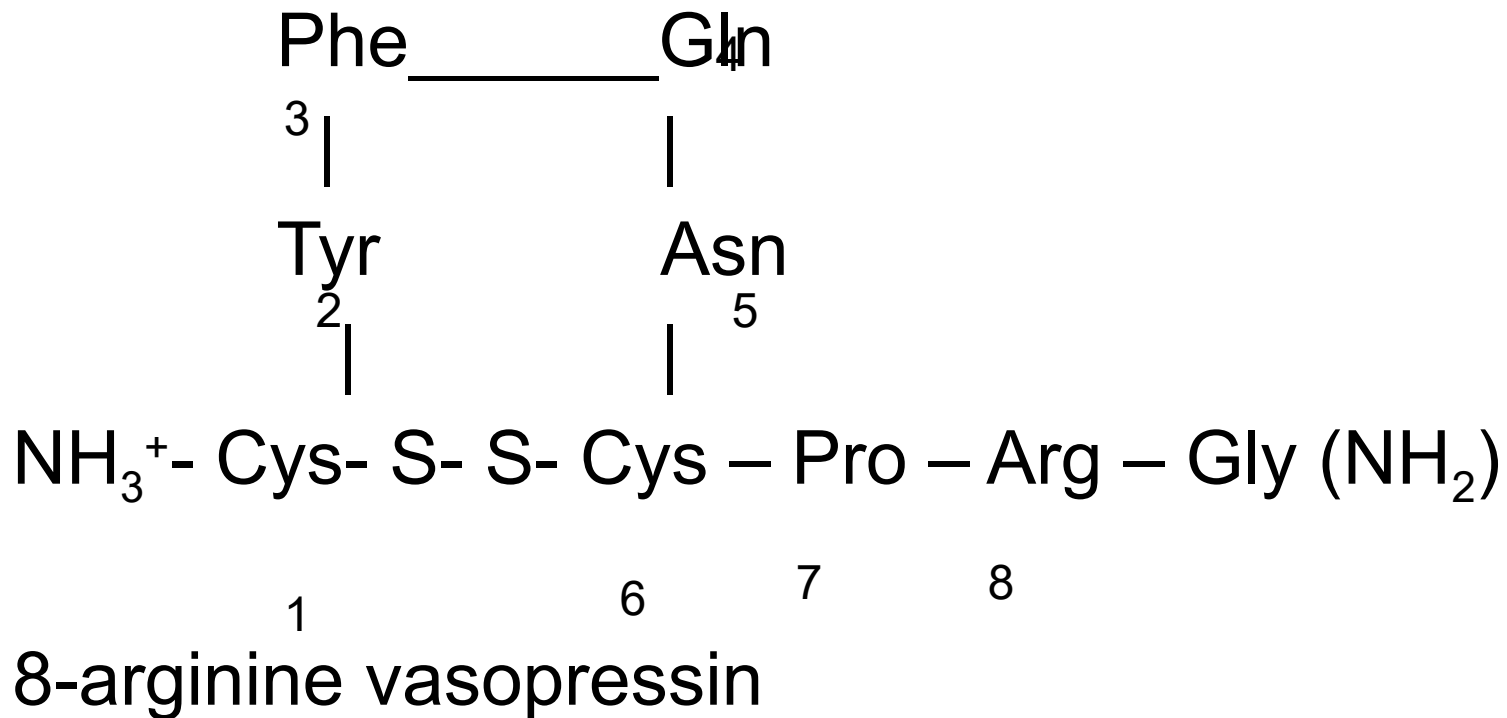
- Vasopressin analogue
- All vasopressin like peptides are nonapeptides,
 - ✓ Contain cysteine residues in positions 1 and 6,
 - ✓ Have an intramolecular disulfide bridge between the two cysteine residues (**essential for agonist activity**),
 - ✓ Have additional conserved amino acids in positions 5, 7, and 9 (asparagine, proline, and glycine, respectively),
 - ✓ Contain a basic amino acid in position 8, and are amidated on the carboxyl terminus.

A) 8-arginine vasopressin

- Also called *vasopressin*, *arginine vasopressin* (AVP), and *antidiuretic hormone* (ADH)

B) Desmopressin 8

- Synthetic peptide and is chemically 1-deamino-8-D-arginine vasopressin
- WHY synthetic analogue???
- ✓ **Increasing duration of action and selectivity for vasopressin receptor**
- ✓ **Less side effect**
- V_2 Selective, **12 times more potent than ADH**
- Also longer duration of action than AVP

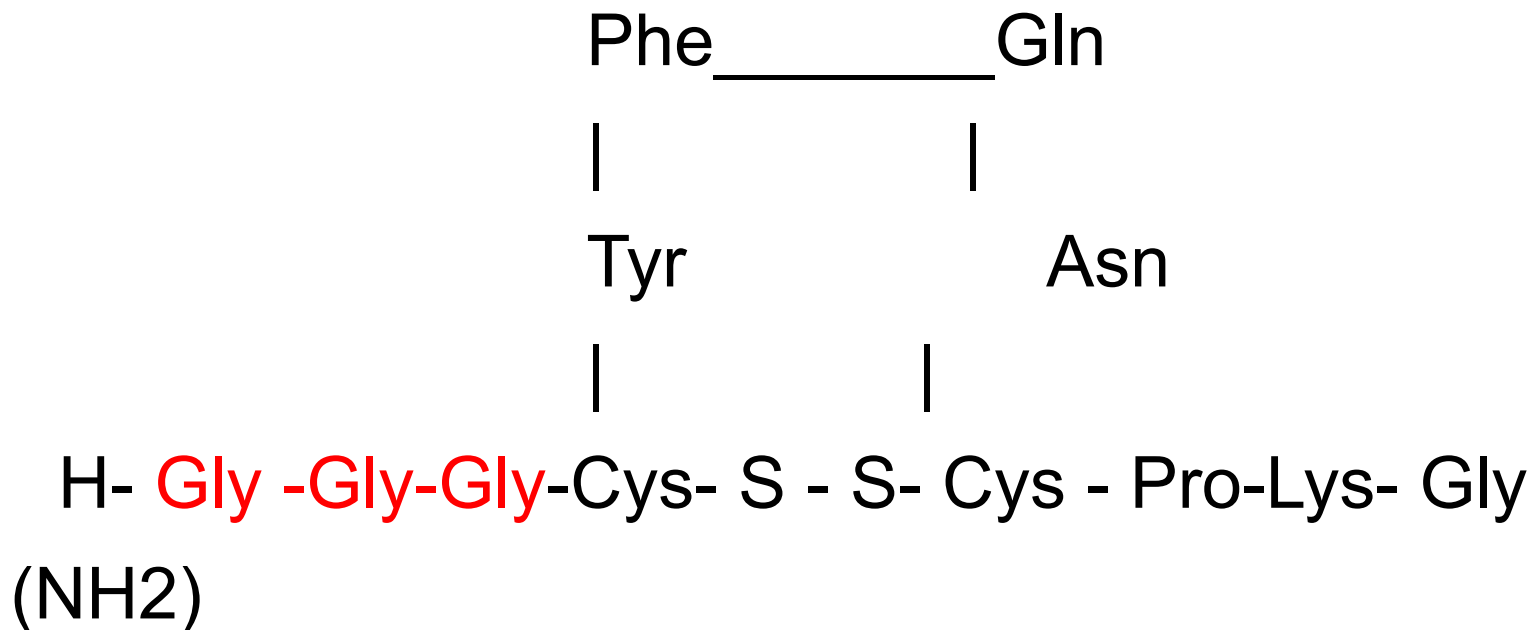


C) Lypressin

- It is 8-lysine vasopressin which act on both V1 and V2 receptors.

D) Terlipressin (triglycyl-lysine vasopressin)

- Synthetic prodrug of 8-lysine vasopressin.



N-triglycyl residue

Therapeutic uses

1. Treatment of diabetes insipidus (DI)
 - DI is disease of **impaired renal conservation of water owing either to an**
 - ✓ Inadequate secretion of vasopressin from the **neurohypophysis** (central DI) or
 - ✓ Insufficient **renal response/resistance to vasopressin** (nephrogenic DI).
 - Characterized by polyuria and polydipsia

Therapeutic uses...

- Central but not nephrogenic DI can be treated with **V₂-receptor agonists** WHY???
 - ✓ **Desmopressin** being the peptide of choice .
 - Vasopressin has little, if any, place in the long-term therapy of DI
 - ✓ Because of **its short duration of action and V₁-receptor-mediated side effects**
2. Bleeding disorders (V₂-mediated, desmopressin), reduce bleeding like in esophageal varices (V₁-mediated, *terlipressin*) ...
3. **Syndrome of Inappropriate Secretion of Antidiuretic Hormone (SIADH).**

Toxicity, Adverse Effects

- Most adverse effects are mediated through the **V₁ receptor acting on vascular and gastrointestinal smooth muscle**
 - ✓ Most serious is **the effect on the coronary circulation**
- The major **V₂-receptor-mediated adverse effect is water intoxication**, which can occur with **desmopressin** or vasopressin.

Reading assignment

Vasopressin receptors antagonists
– Vaptans

Thank you!!